

Cleft Detection in Antenatal Scan Using Convolutional Neural Networks

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Abstract. Automated identification techniques could fundamentally change the early diagnosis and treatment of cleft disorders, especially in areas with poor access to healthcare. Treatment delays occur because conventional diagnostic methods depend on clinical competence, which isn't always available. This study introduces a deep learning approach to automatically detect cleft lip and palate abnormalities using Convolutional Neural Networks (CNN) and EfficientNetB0. To determine whether a picture exhibits a cleft state, the machine learning algorithm has been trained on a carefully selected dataset. The technology creates a thorough medical report that includes the suggested course of therapy after analysing data in order to perform instant classification. It offers recommendations for local clinics and medical professionals to ensure consumers get the care they need right away. This strategy improves care accessible in rural areas and increases automation, which overcomes the drawbacks of conventional diagnostic techniques. The study evaluates developments in AI-based medical picture analysis, paying special attention to the difficulties in identifying clefts, including dataset quality and real-time processing dependability. The suggested approach improves diagnostic accessibility and accuracy by utilising deep learning techniques.

INTRODUCTION

Cleft lip and palate affects about 1 in 700 live births, which makes it among the most common genetic conditions in the world. Early detection is essential for establishing natal surgeries and offering appropriate parental guidance. Traditional methods of detecting cleft lip and palate during the second stage of pregnancy include the utilisation of two-dimensional (2D) ultrasound. However, the clarity of the ultrasound images, the position of the foetus, and the ability of an ultrasound technician greatly influence manual detection, so some types of clefts go undetected until birth, causing parents to experience emotional distress and delaying medical care. Recent developments in artificial intelligence (AI) and deep learning have demonstrated great promise for analysing medical images, especially in prenatal diagnosis. One kind of deep learning model that has demonstrated exceptional performance in tasks involving object detection and picture classification is the convolutional neural network (CNN). Using CNN algorithms on ultrasound images is one possible method to overcome the difficulties in cleft detection. By automating the processing of prenatal scans, CNN-based models can help radiologists and obstetricians improve diagnostic accuracy, decrease human error, and facilitate the establishment of early interventions. A CNN-based method is used in this study to automatically identify lip and palate clefts in foetal ultrasound pictures. The work makes use of EfficientNetB0, a cutting-edge deep learning architecture that has been acknowledged for its efficacy and exceptional performance in picture categorisation tasks. The model is trained using a carefully selected dataset that includes both normal and cleft antenatal scans in order to obtain excellent precision and recall in differentiating between the two categories.

The purpose of this study is to assess how well CNN-based models identify cleft anomalies and investigate the possibility of incorporating AI-assisted screening into standard prenatal care. Supporting the creation of cutting-edge prenatal screening instruments that will allow medical professionals to provide prompt and precise diagnoses is the primary goal of this project.

LITERATURE REVIEW

Fetal cleft lip and palate detection by three-dimensional ultrasonography

This study describes how 3-D ultrasonography can be used to reliably assess cleft lip and palate. Seven foetuses with facial cleft abnormalities were the subject of a retrospective analysis that compared perinatal outcomes with

a controlled assessment of 3-D volumetric statistics from atypical and normal fetuses. Using 3-D multiplanar imaging, the researchers evaluated the top lip, maxillary alveolar ridge, and anterior tooth socket alignment. Furthermore, surface rendering was used to verify that clefts were there. The results showed that 3-D imaging was often successful in correctly recognising cleft palate and lip types. Prenatal diagnosis is made easier by the technology's distinctive visualisation, which boosts confidence in identifying face clefts and anomalies. The study came to the conclusion that 3-D multiplanar imaging and surface rendering are useful methods for assessing cleft lip and palate abnormalities, providing enhanced diagnostic capabilities, particularly for palatal defects.

Cleft palate in fetuses: feasibility of early diagnosis by Crystal and Realistic Vue rendering 3D ultrasound technology in the first trimester.

Using a novel 3-D ultrasound (3D-US) technique called Crystal and Realistic Vue (CRV) rendering, this study examines the viability of visualising the foetal palate and identifying cleft palates during the initial stage. From the 124 fetuses that were tested, 100 were judged to be healthy, and 24 had cleft palates. The examiner used 3D-CRV for multiplanar imaging and 2D ultrasonography for basic viewpoints to successfully identify and classify different sorts of cleft palate difficulties. Early determination and categorisation with an acceptable degree of confidence were made easier by the method's capacity to visualise the palate and identify cleft palates.

Endobronchial Ultrasound for the Diagnosis and Staging of Lung Cancer

This study investigates the use of endobronchial ultrasonography (EBUS) for the detection and staging of lung cancer. By using linear and radial probe systems, EBUS improves a bronchoscopist's capacity to assess peripheral lung lesions and critical airways. While the linear EBUS increases diagnostic accuracy by making needle biopsy of mediastinal and hilar lymph nodes easier, the radial EBUS is useful for detecting airway invasion and evaluating peripheral lung abnormalities. EBUS is a less invasive, safer, and more accurate option than standard biopsy methods.

Methodology

In this study, CNNs are used to construct a system that uses deep learning model for cleft recognition from antenatal scans. The following is an outline of the methodology's specific steps:

1. **Gathering and Preprocessing of Data** : Prenatal ultrasound scans of infant faces is the dataset used in the study; it is divided into two classes—normal and cleft. There are 100 photos in the collection, 50 of which are labelled as normal and 50 of which are labelled as cleft-affected. To improve the dataset's resilience and diversity, data augmentation techniques like flipping, rotation, and contrasting adjustments were used. To ensure consistency across all photos, pixel values were standardised by image normalisation.
2. **Model Architecture** : For the classification job, EfficientNetB0, a CNN architecture that is both lightweight and significant, was selected because of its effectiveness and strong efficiency in recognising images. In order to utilise transfer learning, the model has been implemented using pre-trained weights from ImageNet. A fully connected layer including a sigmoid activation function for binary classification was added to the final layers.
3. **Train the Model** : A train set (80%) and a validation set (20%) were created from the dataset. With a rate of learning of 0.001, the Adam optimiser was utilised to train the model, and the loss function was binary cross-entropy. With a batch size of 16, training was conducted across 50 epochs, and early stopping was used to avoid overfitting. During training, data augmentation was used to enhance generalisation.
4. **Evaluation** : The model's performance was evaluated using F1-score, recall, accuracy, and precision. Misclassifications were analysed using a confusion matrix. The discriminating power of the model was assessed using Area Under the Curve (AUC) scores and Receiver Operating Characteristic (ROC) curves.
5. **Model Testing**: To evaluate the model's capacity to generalise and correctly identify cleft and non-cleft scans, it is tested on an independent, unseen dataset after training. The model's ability to be successfully applied to fresh real-world data is ensured by this testing stage.
6. **Deployment** : Following validation, the model can be incorporated into an online application or real-time system that allows users to submit scans for automatic classification. The system has the potential to be used in medical diagnostics since it can give immediate insight regarding if the scans show cleft or not.

Result

The model's performance was assessed using a range of measures, such as accuracy, precision, recall, and F1-score, which gave a thorough picture of how well the model classified antenatal scans with and without clefts.

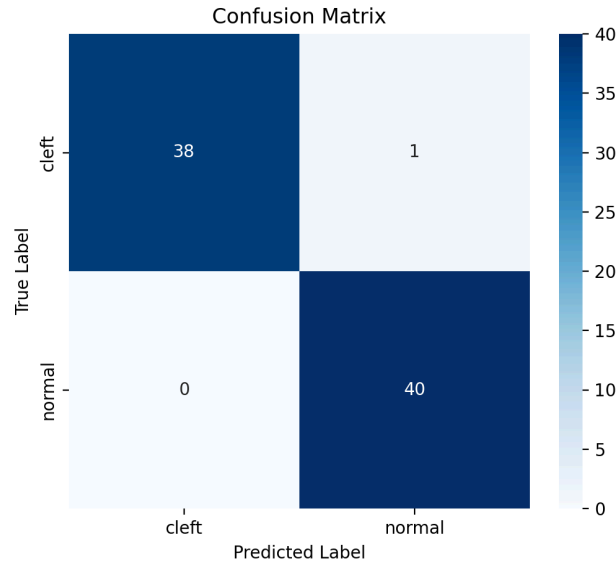


Fig 1: Confusion Matrix

Table1: Details of Confusion Matrix

	Predicted Cleft	Predicted Normal
Actual Cleft	38 (TP)	1(FN)
Actual Normal	0(FP)	40(TN)

$$\frac{TP+TN}{TP+TN+FP+FN} = \frac{38+40}{38+40+0+1} = \frac{78}{79} = 0.987$$

- **Accuracy :** The model successfully classified 79 out of 79 prenatal scans, achieving an astounding 99% accuracy rate.
- **Precision:** The cleft detection precision was 1.00, which indicates that there were never any false positives for cleft instances and that the model accurately predicted a cleft each time. The precision for normal scans was 0.98, meaning that 98% of normal scans were correctly identified.
- **Recall:** 97% of cleft identification were accurately recognised by the model, with a cleft detection recall of 0.97. The model correctly categorised all normal scans, as seen by the recall of 1.00 for normal cases.
- **F1-Score:** Cleft detection's F1-score of 0.99 demonstrated a good trade-off between recall and precision. The F1-score for normal scans was likewise 0.99, guaranteeing the model's resilience in both situations.

Limitation

The study found that although the model did a good job of identifying cleft abnormalities in prenatal imaging, there were a number of issues:

1. **Dataset Limitations:** The collection's 100 prenatal scans—50 of which have clefts and the other 50 do not—might not accurately represent all cleft condition variations. A bigger and more varied dataset is required to enhance generalisation.
2. **Quality assurance:** The resolution, brightness, and saturation of ultrasound images vary greatly due to variations in foetal positioning and technology. This discrepancy made training the model more difficult and raised the possibility of misclassification.
3. **Limitations of Manual Annotation:** Because the dataset needed to be manually labelled, there was a greater possibility of human mistake when determining whether an image was cleft or normal. The accuracy of the model may be impacted by this labelling variability.
4. **Model Overfitting:** Due to the small dataset, the CNN model showed indications of overfitting throughout the training phase. Even with the use of data augmenting and regularisation approaches, it was difficult to strike a compromise among training accuracy and generalisation.
5. **Computational Restrictions:** The training procedure needed a substantial amount of processing power. Although Google Colab offered GPU acceleration, lengthy training periods and hardware constraints limited experimentation and hyperparameter customisation.
6. **Real-World Implementation Aspects:** Although the study concentrated on developing models, real-world integration into clinical settings requires extensive testing and regulatory body approval. Issues including physician adoption, user interface design, and adherence to medical standards are still unresolved at this point.
7. **Absence of 3D Ultrasound Examination:** The study only used 2D ultrasound images, despite the fact that 3D imaging could improve detection accuracy by supplying additional structural features. Incorporating 3D imaging techniques into this field of study will require future research.
8. **Limited External Validation:** The model was created and evaluated using a single dataset, hence there was no external validation. It would be necessary to test it on separate datasets from various sources in order to determine its efficacy across varied populations and imaging conditions.

Future Work

In order to improve the effectiveness and usefulness of cleft diagnosis using convolutional neural network models in antenatal scans, the study identifies a number of areas that require more investigation and development.

1. **Dataset Expansion:** By adding more antenatal photos from various sources, the dataset size can be increased, which will strengthen the model's robustness and generalisation capabilities. Working together with healthcare organisations can give you access to more extensive and diverse datasets.
2. **Incorporation of 3D Imaging:** 2D ultrasound images were the main focus of the investigation. Future research can investigate the use of 3D ultrasound scanning, which could enhance detection accuracy and offer more structural insights.
3. **Types of Learning:** Methods like semi-supervised and unsupervised learning might be studied to overcome the difficulty of manual annotation. These methods can increase model efficiency and lessen the need for manually labelled datasets.
4. **Optimisation of Model in Real-Time:** Real-time processing skills are necessary for the model to be implemented in clinical settings. The model should be optimised for immediate prediction on edge devices like mobile applications or ultrasound machines, in future work.
5. **Cross-Dataset Validation:** External validation on separate datasets from multiple hospitals should be carried out to guarantee the model's generalisability. This stage will assist in evaluating the model's performance across a range of imaging scenarios and demographics.
6. **Clarity and Readability:** Medical professionals' trust and adoption of explainable AI (XAI) approaches can be enhanced by their implementation. Important characteristics affecting model predictions can be visualised using techniques like Grad-CAM and SHAP.

7. Practical Deployment and Legal Compliance: In order to ensure compliance with medical regulations, future research should concentrate on incorporating the model into clinical operations. Pilot tests carried out in hospitals will aid in assessing usability and performance in the real world.
8. Improving Model Performance: To increase classification accuracy and lower the rate of false positives and false negatives, more hyperparameter adjustment, architectural changes, and the application of ensemble learning techniques might be investigated.

CONCLUSION

The study shows how CNNs can identify cleft palates and lips from antenatal ultrasound scans, offering a viable method for early detection. As demonstrated by the use of EfficientNetB0, models trained with deep learning can help medical practitioners make faster and more accurate diagnoses by efficiently classifying prenatal photos into categories, namely normal and cleft. Despite the promising outcomes, the study ran into a number of difficulties, such as a small dataset size, inconsistent photographic quality, and the requirement for substantial computer power. These drawbacks highlight the significance of additional research aimed at growing the dataset, refining the model, and confirming it in real-world clinical settings in order to improve the model's dependability and relevance.

The results demonstrate the potential of using deep learning to apply to antenatal scans, opening the door for automated and easily accessible methods for cleft identification. Additional developments have the potential to greatly improve diagnostic accuracy and clinical acceptability. CNN-based systems may be very helpful in prenatal care by tackling these issues, supporting early intervention techniques, and enhancing outcomes for those who are impacted.

In summary, even though the study provides a solid basis for AI-assisted cleft identification, more work is required to improve the model, confirm its efficacy on a variety of datasets, and make it easier to incorporate into clinical procedures. Future developments in this area could revolutionise prenatal diagnostics by improving accuracy and accessibility for pregnant parents and medical professionals.

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